

EE 542: Modelling, Simulation and Control of Hybrid Electric Vehicle

Course Code	EE 542
Course Name	Modelling, Simulation and Control of Hybrid Electric Vehicle
L-T-P-C	3-0-0-3
Prerequisites	Linear Algebra
Intended for	3rd and 4th year UG, PG, PhD
Approval	50th BoA (Jul 2023)

Course Contents

- **Modelling in performance parameter:** Modelling Vehicle Acceleration-Acceleration performance parameters, modeling the acceleration of an electric scooter, modeling the acceleration of a small car. (5 hours)
- **Modelling of Battery Electric Vehicles:** Electric Vehicle Modelling Tractive Effort, Rolling resistance force, Aerodynamic drag, Hill climbing force, Acceleration force, Total tractive effort, Modelling Electric Vehicle Range-Driving cycles, Range modeling of battery electric vehicles, Constant velocity range modelling, Range modelling of fuel cell vehicles, Range modelling of hybrid electric vehicles. (8 hours)
- **Drive Train Characteristics:** Modelling and Characteristics of EV/HEV Power trains Components-ICE Performance Characteristics, Electric Motor Performance Characteristics- Battery Performance, Characteristics-Transmission and Drive train Characteristics- Regenerative Braking Characteristics-Driving Cycles Modelling and Analysis of Electric and Hybrid Electric Vehicles Propulsion and Braking-Longitudinal Dynamics Equation of Motion-Vehicle Propulsion Modelling and Analysis-Vehicle Braking Modelling and Analysis. (8 hours)
- **Energy Management:** Handling Analysis of Electric and Hybrid Electric Vehicles-Simplified Handling Models Energy/Power Allocation and Management-Power/Energy Management Controllers-Rule-Based Control Strategies- Optimization-Based Control Strategies. (8 hours)
- **Vehicle Dynamic Control:** Control of Electric and Hybrid Electric Vehicle Dynamics-Fundamentals of Vehicle Dynamic Control (VDC) Systems, VDC Implementation on Electric and Hybrid Vehicles Case Studies, Rechargeable Battery vehicles, Hybrid Vehicles, Fuel Cell Powered Bus. (8 hours)
- **Estimation Techniques:** Identification of important state variables and parameters of HEV, Kalman filter-based methods, Least Square based methods. (5 hours)

Text books: (Relevant and Latest, only 2)

1. Wei Liu, **Introduction to Hybrid Vehicle System Modeling and Control**, Wiley, 2015.
2. Y. Xu, J. Yan, H. Qian, and T. L. Lam, **Hybrid Electric Vehicle Design and Control**, vol. 1, McGraw Hill, 2021.

References

1. James Larminie, John Lowry, **Electric Vehicle Technology Explained**, John Wiley & Sons Ltd, 2003.
2. Amir Khajepour, Saber Fallahand Avesta Goodarzi, **Electric and Hybrid Vehicles-Technologies, Modelling and Control: A Mechatronic Approach**, John Wiley & Sons Ltd, 2014.
3. Antoni Szumanowski, **Hybrid Electric Power Train Engineering and Technology: Modelling, Control, and Simulation**, IGI Global, 2013.
4. Mehrdad Ehsani, Yimin Gao, Ali Emadi, **Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design**, 2nd Edition, CRC Press, 2010.